

How to connect to CloudSQL DB using IAM service account from application running on local machine

Pre-Requisite

1. Service Account is added as IAM user on the CloudSQL instance.
2. Service account has the required roles and permissions to connect to the database, schema and tables and perform CRUD operations.
3. You already have Service Account's json key.

Install Cloud SQL Auth Proxy

macOS (Homebrew)

```
brew install cloud-sql-proxy
```

Linux (Debian/Ubuntu)

```
curl -o cloud-sql-proxy https://dl.google.com/cloudsql/cloud-sql-proxy.linux.amd64
chmod +x cloud-sql-proxy
sudo mv cloud-sql-proxy /usr/local/bin/
```

Windows (PowerShell)

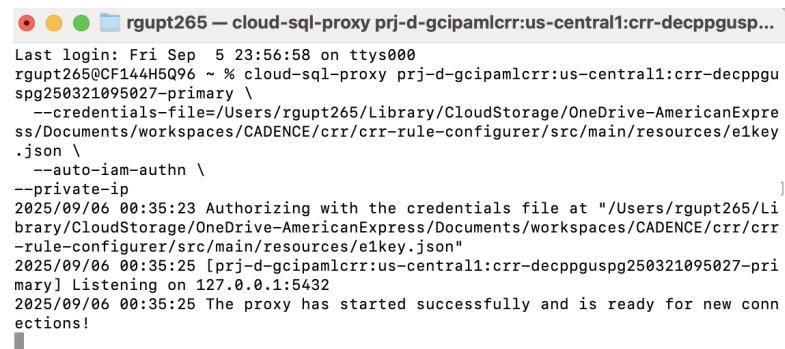
Download from official release page and put it in PATH.

Run the Proxy

Once installed, run (replace with your values):

```
cloud-sql-proxy <PROJECT_ID>:<REGION>:<INSTANCE_NAME> \
  --credentials-file=/path/to/service-account.json \
  --auto-iam-authn \
  --private-ip
```

Something like below should come after successful run



```
rgupt265 — cloud-sql-proxy prj-d-gcipamlcrr:us-central1:crr-decpggusp...
Last login: Fri Sep  5 23:56:58 on ttys000
rgupt265@CF144H5Q96 ~ % cloud-sql-proxy prj-d-gcipamlcrr:us-central1:crr-decpggusp250321095027-primary \
  --credentials-file=/Users/rgupt265/Library/CloudStorage/OneDrive-AmericanExpress/Documents/workspaces/CADENCE/crr/crr-rule-configurer/src/main/resources/e1key.json \
  --auto-iam-authn \
  --private-ip
2025/09/06 00:35:23 Authorizing with the credentials file at "/Users/rgupt265/Library/CloudStorage/OneDrive-AmericanExpress/Documents/workspaces/CADENCE/crr/crr-rule-configurer/src/main/resources/e1key.json"
2025/09/06 00:35:25 [prj-d-gcipamlcrr:us-central1:crr-decpggusp250321095027-primary] Listening on 127.0.0.1:5432
2025/09/06 00:35:25 The proxy has started successfully and is ready for new connections!
```

This will listen on your laptop at:

- Postgres localhost:5432
- MySQL localhost:3306

Update Spring Boot Config

Postgres

```
spring.datasource.url=jdbc:postgresql://localhost:5432/<DB_NAME>
spring.datasource.username=<IAM Service Account whose key json was used to run the proxy>
spring.datasource.password=
```

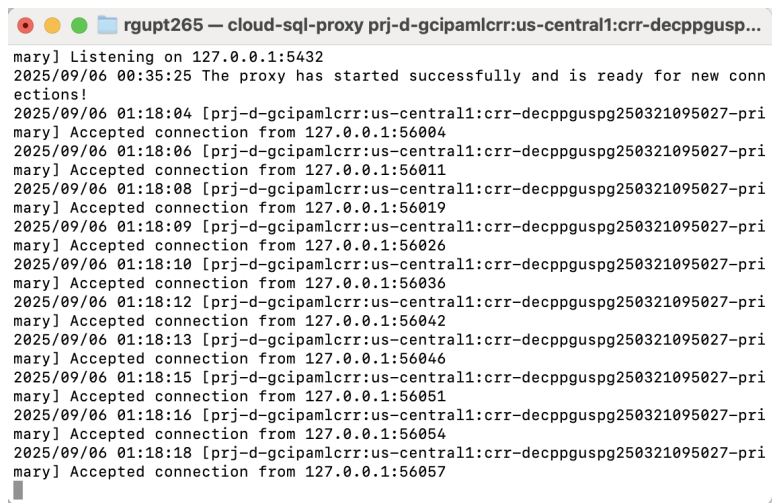
MySQL

```
spring.datasource.url=jdbc:mysql://localhost:3306/<DB_NAME>
spring.datasource.username=<IAM Service Account whose key json was used to run the proxy>
spring.datasource.password=
```

Run Spring Boot Application

Make sure cloud sql proxy is running in one terminal before you start your spring boot application.

Spring boot application will get seamless db connections via the proxy.

A terminal window titled "rgupt265 — cloud-sql-proxy prj-d-gcipamlcrr:us-central1:crr-decpggusp..." displays the logs of the cloud-sql-proxy. The logs show the proxy listening on 127.0.0.1:5432 and successfully accepting multiple database connections from the same address. The timestamp for all log entries is 2025/09/06 01:18:04 through 01:18:18.

```
mary] Listening on 127.0.0.1:5432
2025/09/06 00:35:25 The proxy has started successfully and is ready for new conn
ections!
2025/09/06 01:18:04 [prj-d-gcipamlcrr:us-central1:crr-decpgguspg250321095027-pri
mary] Accepted connection from 127.0.0.1:56004
2025/09/06 01:18:06 [prj-d-gcipamlcrr:us-central1:crr-decpgguspg250321095027-pri
mary] Accepted connection from 127.0.0.1:56011
2025/09/06 01:18:08 [prj-d-gcipamlcrr:us-central1:crr-decpgguspg250321095027-pri
mary] Accepted connection from 127.0.0.1:56019
2025/09/06 01:18:09 [prj-d-gcipamlcrr:us-central1:crr-decpgguspg250321095027-pri
mary] Accepted connection from 127.0.0.1:56026
2025/09/06 01:18:10 [prj-d-gcipamlcrr:us-central1:crr-decpgguspg250321095027-pri
mary] Accepted connection from 127.0.0.1:56036
2025/09/06 01:18:12 [prj-d-gcipamlcrr:us-central1:crr-decpgguspg250321095027-pri
mary] Accepted connection from 127.0.0.1:56042
2025/09/06 01:18:13 [prj-d-gcipamlcrr:us-central1:crr-decpgguspg250321095027-pri
mary] Accepted connection from 127.0.0.1:56046
2025/09/06 01:18:15 [prj-d-gcipamlcrr:us-central1:crr-decpgguspg250321095027-pri
mary] Accepted connection from 127.0.0.1:56051
2025/09/06 01:18:16 [prj-d-gcipamlcrr:us-central1:crr-decpgguspg250321095027-pri
mary] Accepted connection from 127.0.0.1:56054
2025/09/06 01:18:18 [prj-d-gcipamlcrr:us-central1:crr-decpgguspg250321095027-pri
mary] Accepted connection from 127.0.0.1:56057
```